

Space–Time–Behavior Flow-Through Tensors: A Certified Multi-Axis Operator Framework Connecting Logit SUE, Fluid Queues, and Layered Network Equilibria

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Xuesong Zhou [co-authors TBD]

Draft v0.6 — every number reproduced from the certified test suite (49 tests) and run scripts of `stb_ftt_unified_v0.6`

Abstract

The Flow-Through Tensor (FTT) architecture connects origin–destination flows, path probabilities, and link travel times as chained tensors on one computational graph. This paper wraps that foundation into a *certified multi-axis operator framework*: every axis of the space–time–behavior tensor declares exactly a *status* (spectator, contracted, or synchronized), a *semiring* $((+, \times)$ flow, $(\min, +)$ pricing, $(\max, \times)$ state decode), and a methodological *anchor*; a model is a chain of promotions of axes from spectator to synchronized; and every synchronized axis owes one well-posedness argument and one certificate class. Three foundational special cases close the framework from below — logit stochastic user equilibrium and its deterministic limit (behavior), Newell fluid point queues (time), and the flow-through tensor chain itself (space) — each verified against closed forms inside one code base. Above them, the same operator calculus carries multi-layer passenger–freight price consensus, the passenger–vehicle service coupling $Rx \leq Sy$, and a mixed-autonomy ride-hailing profile with competing companies, AV/HV fleets, and patience-capped matching queues. A solver-engine escalation ladder (damped fixed point $\rightarrow$ averaging $\rightarrow$ Anderson $\rightarrow$ stiff-block Newton $\rightarrow$ semismooth NCP/VI) computes the fixed points that pin synchronized axes, with cycle detection and engine bookkeeping as part of the certificate. All claims are bounded by an explicit validity domain: learned or behavioral terms are flow-independent along synchronized axes, compression is admissible only on column-rich spectator axes, and nonsmooth dynamic loading is a stated physics boundary, not a hidden smoothing.

1 Introduction

Three research lines meet in this paper. First, the FTT computational graph, which chains $\mathbf{f}_{OD} \rightarrow \mathbf{f}_P \rightarrow \mathbf{f}_L \rightarrow \mathbf{t}_L \rightarrow \mathbf{t}_{OD}$ through typed incidence operators and exploits the transpose symmetry of contraction for gradients. Second, the classical equilibrium canon — logit SUE, deterministic UE, and fluid queue loading — whose recovery as special cases is the ground truth any unifying framework must pass. Third, layered service systems — multi-commodity freight, passenger–vehicle service coupling, and mixed-autonomy ride-hailing — whose defining feature is that layers interact *only* through shared resources synchronized by prices.

The claim of this paper is deliberately narrow, in the sense fixed by our earlier position notes: we do not claim a new three-way unification of neural, optimization, and transportation architectures. We claim a *certified operator framework*: (i) an axis calculus in which the classical models are exact

reductions, verified numerically to closed-form accuracy; (ii) a scalable column-generation layer whose optimality gaps are always priced over the full network, never over the enumerated pool; (iii) a layered-equilibrium layer whose prices are certified against exact duals where the coupling is convex, and whose complementarity conditions are audited where it is not; and (iv) an explicit engine ladder for the fixed points, with failure modes diagnosed rather than tuned away.

2 The axis calculus

Let the demand tensor carry indices $(o, d, g, m, \tau, p, a, t)$ — origin, destination, group, mode, departure period, path/column, link, entry time — extended by profile-specific axes (company, vehicle type, operational stage, matching queue). Every axis declares exactly three properties:

declaration	values	meaning
status	spectator / contracted / synchronized	rides along / summed out / pinned by a fixed point
semiring	$(+, \times)$, $(\min, +)$, $(\max, \times)$	flow aggregation / pricing / state decode
anchor	$S_{\text{repr}}, S_{\text{stat}}, S_{\text{mech}}, S_{\text{ctrl}}$	which methodological layer owns it

Four consequences follow mechanically.

Models are promotion chains. Static UE promotes only the link axis (congestion fixed point). Dynamic assignment additionally promotes departure/entry time (queue conservation recursion). Multi-layer systems promote the layer axis (price consensus). State estimation promotes the regime axis under $(\max, \times)$. The mixed-autonomy profile promotes the matching-queue axis (patience-capped waiting).

Compression lives on spectator axes only, and only where columns are rich. The Chicago Sketch experiment below shows the boundary empirically: at $\bar{K} \approx 2.2$ active columns per OD, route-family atoms cannot pay.

Learning is admissible only if flow-independent along every synchronized axis (the clean-VI boundary): priors, values, and utility offsets may be learned; anything entering a flow-dependent cost voids the certificates unless monotone by construction.

Each synchronized axis owes one certificate class: monotonicity $\Rightarrow$ FW/KKT gap; causality $\Rightarrow$ conservation and closed-form pulses; convex duality $\Rightarrow$ consensus gap against exact duals; integrality or nonmonotone coupling $\Rightarrow$ report the residual gap, never assume it away.

3 Special-case closure

All three foundational cases are certified inside one package (`tests/test_v06_closure.py` and the Sioux Falls special-case suite; data: TCGLite GMNS Sioux Falls, 24 nodes / 76 links / 33 ODs / 126 behavioral paths).

3.1 Behavior: logit SUE and the deterministic limit

The entropy-regularized master over the fixed path pool has, at frozen costs, the multinomial logit as its *exact* KKT solution (machine precision, 0.0 error). With congestion, Evans-type partial linearization with exact line search yields self-consistency residuals $\leq 2 \times 10^{-5}$ across $\theta \in [0.1, 50]$, and the Wardrop gap falls monotonically from 6.9×10^{-2} at $\tau=10$ to 8.3×10^{-6} at $\tau=0.02$: the $\tau \rightarrow 0$ ladder reaches deterministic UE inside one solver family. Two solver lessons are part of

the record: the plain damped fixed point *oscillates* at high congestion (residual 0.99), and demand scaling matters ($\times 25$ gives $v/c = 4.7$ and an ill-conditioned master; $\times 8$, $v/c \approx 2$, is the certified stress level).

A full-network Dijkstra audit prices the fixed 126-path pool at every reported solution: the pool is *exact* at free flow (relative gap 1.9×10^{-16}) and *35.7% inadequate* at congested UE times. The static-pool failure mode is therefore measured, not assumed, and adaptive column generation is the certified remedy.

3.2 Time: Newell fluid point queues

The point-queue recursion reproduces the closed-form constant-overload pulse ($\lambda=30$, $\mu=20$, $\tau_0=10$ min) to relative errors $\leq 7 \times 10^{-5}$ in peak queue, clearing time, and total delay; network flow-through of the certified UE column flows conserves mass to 9×10^{-12} vehicles over 71 loaded links, with causality $D(t) \leq A(t)$ everywhere. The boundary is stated with the operator: sending/receiving $\min(\cdot)$ is nonsmooth at binding capacity; smoothing restores gradients but changes the physics; no differentiability through the queue is claimed.

3.3 Space: the flow-through tensor chain

The TCG chain $\alpha \rightarrow \Gamma \rightarrow P \rightarrow \Delta$ (zone generation, OD split, path proportion, link incidence) is exactly the conserved decoder composition $x = D(\Pi)X$: row-stochastic masked layers, mass conservation 2×10^{-13} , link reconstruction residual $< 10^{-9}$, and the forward chain at target observations reproduces the sensor counts to the data’s own rounding (max relative error 1.0×10^{-2}). The calibration direction carries an analytic gradient through the masked-softmax chain, verified by finite differences to 2×10^{-7} — with the methodological caveat that the first FD check was vacuous (masked coordinates have identically zero gradient and difference) until sampling was restricted to meaningful coordinates.

4 Scale and the compression boundary

The same solvers run unchanged on a deterministic grid generator, Sioux Falls, and Chicago Sketch (933 nodes, 2,950 links, 386 zones, 93,135 ODs, 1.14M trips) through one canonical GMNS contract. Full-network certified results (single process, single thread; all gaps priced by all-origin Dijkstra over the real network):

algorithm	full-space rel. gap	wall (s)	columns	atoms
CG Frank–Wolfe	8.3×10^{-5}	7.4	198,871	—
CG FW + grouped-simplex GP	9.4×10^{-5}	39.3	207,248	—
route-family atoms (static)	6.1×10^{-3}	141.9	200,712	157,373
route-family atoms (adaptive)	4.6×10^{-3}	178.4	215,237	220,152

The certified link flows correlate 1.0000 with the network’s reference volumes (RMSE 23.8 veh). The atom rows are the honest boundary datum: at $\bar{K} \approx 2.2$ columns per OD, the compression ratio is only $1.28\times$, adaptive promotion generates 62,761 singletons until atoms *outnumber* columns, and the gap floor sits two orders above tolerance — reproducing, at scale, the crossover law that compression pays only in column-rich regimes (activity hierarchies, transit itineraries), never in the coupling itself. Wall-clock numbers support algorithm ordering only; engine-level speed claims are out of scope.

5 Layers and prices

5.1 Convex two-layer benchmark with exact dual certificates

Passenger columns (mandatory, PCE 1) and elective e-commerce freight requests (PCE 2) share link capacity on Sioux Falls. With linear costs the joint problem is an LP: 11 binding links, max price 40.8 min/veh-equiv. Iterative block coordination — parallel min-plus best responses per layer, projected subgradient on the shared price, primal averaging — certifies against the exact optimum: dual-bound gap 3.3×10^{-4} , price agreement within 1.6% on every binding link, served freight 38.1% falling to 32.4% as slack tightens. Replacing hard capacity by BPR congestion keeps joint convexity (one Beckmann potential in total veh-equivalent load): joint FW gap 1.8×10^{-6} ; Gauss–Seidel block coordination agrees to 5.3×10^{-7} with link loads within 0.12 veh-equiv; freight threshold complementarity is exact; and congestion *alone* prices freight from 100% served ($\times 4$ passenger demand) to 86.3% ($\times 16$) — the congested travel time is itself the synchronizing price. The step-size lesson is recorded: the subgradient step must be scaled to the load/price ratio ($\rho_0=0.5$ diverges; 0.01 converges; report the best dual bound).

5.2 The passenger–vehicle coupling and the integer corner

The service coupling $Rx \leq Sy$ with a scarcity price on service capacity is carried in the package as its own layer, with passenger mass, service capacity, vehicle flow, and queue mass audited separately. The activity–travel vehicle-scheduling line (the “how many trip requests” formulation) is precisely the *integer corner* of this master: passenger chains as columns, one-chain constraints as simplexes, on-board arc coupling as incidence, and the Lagrangian trip price as the synchronizing dual. On the integer side the duality gap need not vanish and must be reported; our convex benchmarks are the certified relaxation of that corner.

5.3 The mixed-autonomy ride-hailing profile

The MAGE class (competing TNCs; AV/HV fleets under an AV-share cap; pickup and service stages; patience-capped customer matching; congestion-coupled routing) enters as an *operator profile*: extension axes (company, vehicle type, stage, matching queue) are registered without touching the canonical registry; the matching queue is kept structurally distinct from the road queue; and the road layer reuses the certified full-space assignment each outer iteration. On an 8×8 grid with two asymmetric companies the equilibrium closes in 4 outer iterations (inner residual 4×10^{-14} ; conservation 10^{-13} ; patience, rationing-complementarity, and fleet-cap violations 0; multi-start share spread 4×10^{-6}). Comparative statics behave economically: raising one company’s AV cap from 0 to 1 moves its AV share from 6.7% to 29.8% and cuts its matching wait from 12.0 to 1.8 minutes; fleet scale $\times 0.3 \rightarrow \times 1.5$ moves served fraction from 46% to 100%; and the patience sweep exposes the defining subtlety — *lower* patience (4 min) *raises* ride-hailing share to 64.8% while service is rationed to 77.9%, because patience caps the disutility, not the supply, with the complementarity (unserved $> 0 \Rightarrow$ wait at patience) holding exactly. The full NCP/VI equilibrium (PATH class) is the declared next solver rung; the profile’s existence claims are restricted to the steady-state operators, exactly as in the source formulation.

6 Equilibrium engines

Fixed points that pin synchronized axes are computed by an explicit escalation ladder — damped Picard (E0), MSA averaging (E1), safeguarded Anderson acceleration (E2), stiff-block Newton

(E3), and semismooth NCP/VI (E4, declared) — with a period- p cycle detector as the promotion trigger and the engine choice recorded in the result. The motivating instance is part of the test suite as an honest negative: the ride-hailing choice $\leftrightarrow$ wait subsystem has $|\partial w/\partial s| \sim 10^4$ near saturation, damped iteration limit-cycles at residual 0.12 (detected, period 4), smoothing and two-timescale damping do not rescue it, and the exact Newton solve of the *smallest stiffest block in its low-dimensional dual-like variable* (the wait vector) closes it. That design rule — solve the stiff block exactly in waits/prices/multipliers, never in high-dimensional primal shares; keep slow blocks in cheap fixed-point form — is the solver-side counterpart of the coordination-vs-synchronization distinction.

7 Roadmap: the phase–time axis (v0.7)

The deferred state/regime axis has a concrete target: the phase-augmented DTA formulation with X (queue dynamics), Y (generalized phase selection), and Z (vehicle trajectory) blocks coupled by Lagrangian relaxation. In the axis calculus this is one further promotion: the regime axis becomes synchronized, contracted under $(\max, \times)$ for decoding and $(\min, +)$ for pricing on the phase-expanded network, with within-phase closed-form queue profiles replacing the binding $\min(\cdot)$ — moving the nonsmoothness to the phase boundary where a discrete engine (E4/branch-and-price) owns it. The engine ladder and the consensus machinery above are exactly the solver substrate this promotion requires.

8 Validity domain and limitations

(i) All learned/behavioral terms in every certified experiment are flow-independent along synchronized axes; the monotonicity boundary is stated wherever a flow-dependent extension would cross it. (ii) The dynamic loading operator is forward-only; no gradients are claimed through binding capacities. (iii) Integer couplings (vehicle tours, phase selection) carry duality gaps that are reported, not assumed zero. (iv) The well-posedness map currently spans behavioral-feedback and queue-feedback strength; the coupling-integrality axis is documented but not yet swept. (v) Wall-clock comparisons support algorithm ordering on one machine, single-threaded; engine-level speed claims require the separate compiled-solver line.

9 Conclusion

The FTT chain supplies the spine; the axis calculus supplies the grammar; the special cases nail the framework to closed forms; the layered benchmarks show that one price mechanism — explicit duals under hard coupling, congested times under soft coupling, patience-capped waits under service coupling — synchronizes every layer pair; and the engine ladder makes the computation of those fixed points a diagnosed, certified choice rather than a tuning exercise. Everything above is reproducible from one package with one test suite (49 tests) and the run scripts cited throughout.

Reproducibility. `python -m pytest (49 tests); run_mage_grid.py, run_unified_grid_harness.py, run_passenger_vehicle_harness.py` in `stb_ftt_unified.v0.6`; multi-layer benchmarks in `STB_multilayer_bench` (7 tests); special-case artifacts in `STB_special_cases_SF` (7 tests). [\[Bibliography, author list, and venue formatting pending author decisions; FTT manuscript and MAGE submission to be cited as companion/related work.\]](#)